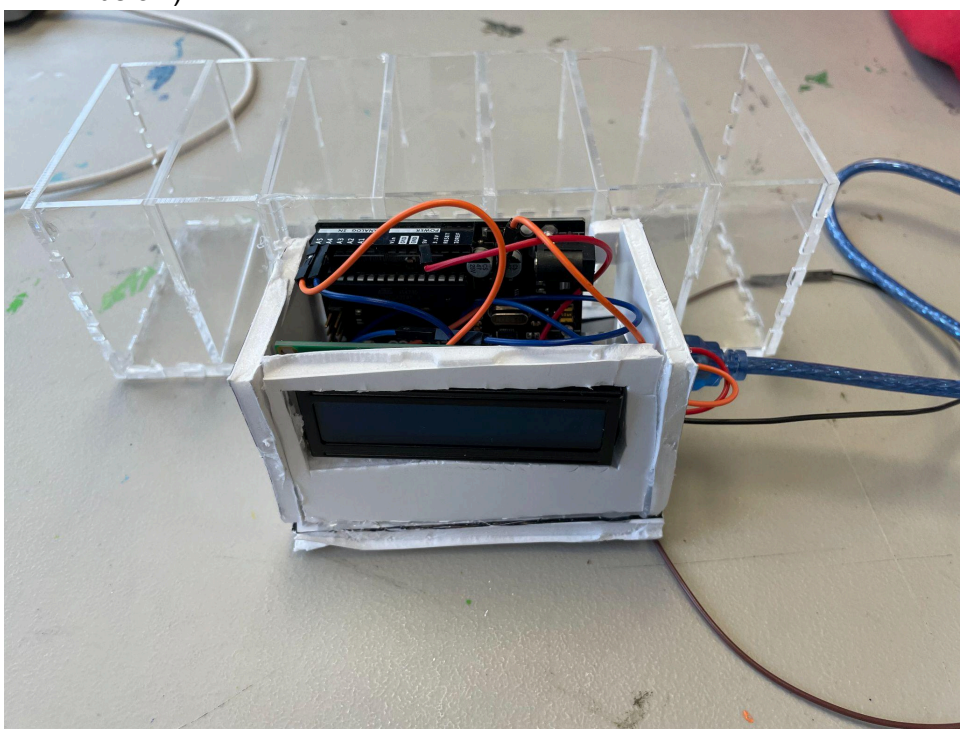
This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

1. For our final solution, we decided to make a pill box that has 7 different slots for each day of the week: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday. We required that it had to be big enough to fit all of the user's medication, that it had to notify the user via a displayed message on the LCD, and that the mechanics must be able to fit onto the box to optimize space. For our final prototype we had a case for the Arduino on the front of the box, a LCD display on the front, and a pill box (see picture below).

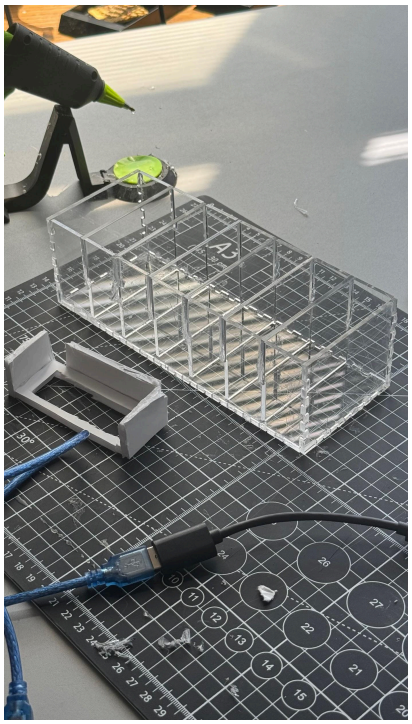
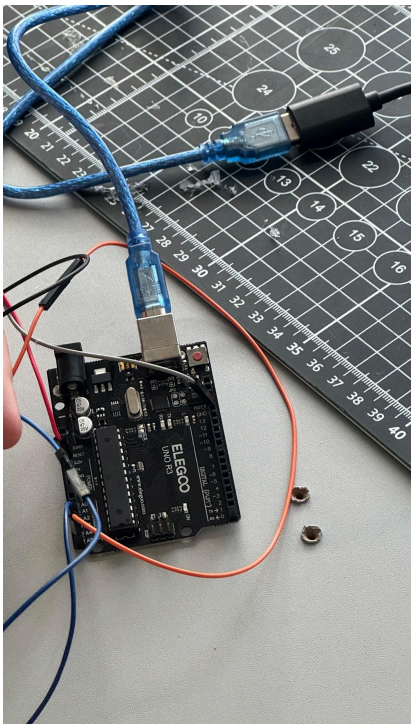
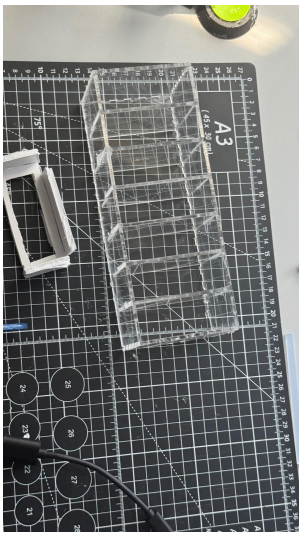


2. Our plan of action was to first go into the makercase and develop a box that had 7 columns where you could put each day's medication in and laser cut it. The second was to make the container for the motherboard and LCD monitor and make sure we had a hole on the side where the door hinge sensors and ground wire can come out of. The third was to laser cut a lid for the pill box and attach the magnetic hinge to it and implement the code for the door sensors into the arduino app.

3. We began the process of generating a solution by thinking of the problems that we could solve by creating a pill box that notifies the user when to take their pills. We originally thought of an app but decided on a display screen because that way you cannot silence

the notification and it will show up. It also requires a more simple set up. A display is more accessible because it does not require the user to download an app. Something we would add to our prototype would be sections for different parts of the day. Most medications are not taken at the same time. The implementation of at least three different sections for each day of the week would be very helpful. One morning, afternoon and before bed.

Plan:

1. Measure out how big you want your pill box to be, 4-4-7 inches.
2. Create a box plan with hinges and 7 compartments to match the 7 days of the wee,k and a designated section to put the motherboard and display screen.
3. Assemble the ox.

We assembled using notches and hot glue. This had no cost to us because all the supplies were provided by the school. If someone wanted to recreate this, [the MakerCase website]() was free to develop and download the box plans. Some stores like Laser Works, KW Tek Laser Products, and Innovative Laser Design will laser cut items for you, and you can usually find hot glue guns and hot glue sticks at Office depot and other office supply stores.